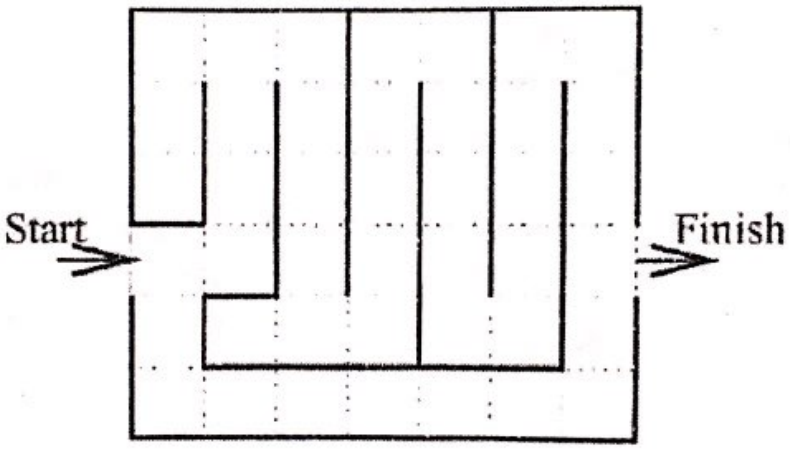
Faculty of Science
Department of Mathematics
Comp 408 / Final exam / June 2011
Four pages

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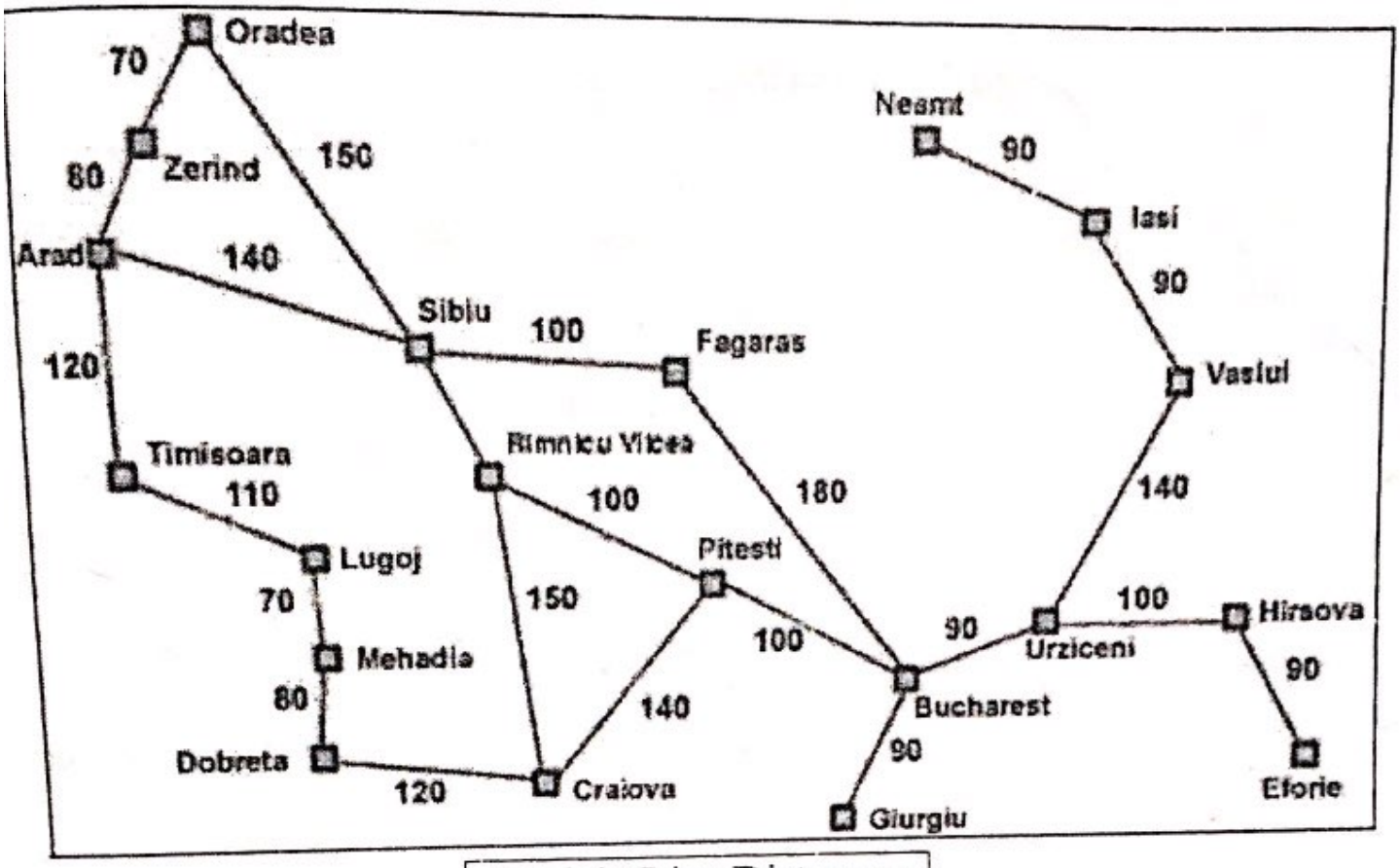
1. [10 marks] An informed search algorithm can easily solve this puzzle, getting from the start location to find the finish location (see figure below). The operators are up, down, left, right, which are only valid if there is a dashed line between the grid boxes in the map of the puzzle. Consider that the heuristic used is the straightline distance between the current position in the puzzle and the exit from the puzzle.



   a. Show that hill-climbing search will fail to find an optimal solution. (Your answer should make it clear why it fails to find the optimal solution).
   b. Which informed search algorithm would guarantee finding the optimal solution?

2. [10 marks] Consider the problem of driving from Arad to Bucharest (figure below). The map show driving distances between cities and the table lists straight line heuristic distances from selected cities to Fagaras. Draw the search tree for an A* strategy applied to finding the optimal route from Vaslui to Fagaras. Put the city name and f (n) value in each node, and driving distance beside each arc. Show all the nodes that are generated.

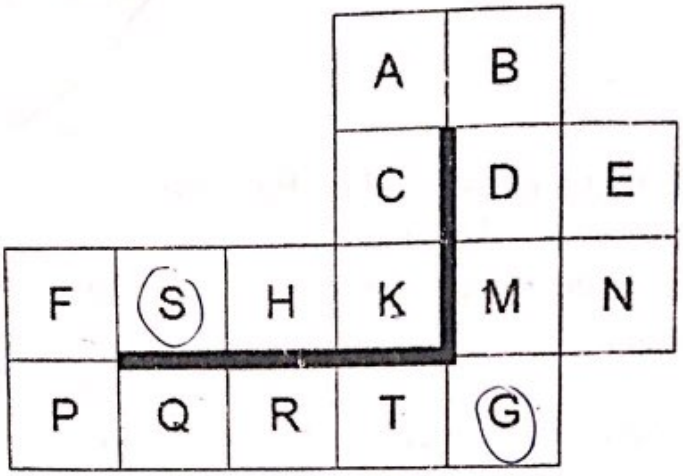
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Straight Line Distances to Fagaras	
Bucharest	160
Craiova	180
Eforie	310
Fagaras	0
Giurgiu	200
Hirsova	260
Iasi	180
Neamt	140
Pitesti	90
Rimnicu Vilcea	90
Urziceni	190
Vaslui	220

3. [26 marks] Consider the following puzzle in which the successors of a cell include any adjacent cell in the directions North, South, East, and West of the current cell, except at the boundary of the puzzle or when a barrier (thick line) exists. For example, successors (M) = {D, N, G}. Assume each move has cost 1.

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The problem is to find a path from cell S to cell G. What is the order of nodes expanded (plus the goal node if it is found) by each of the following search methods?

a. Depth-First Search. Assume cycles are detected and eliminated by never expanding a node containing a state that is repeated on the path back to the root.
b. Greedy Search. Use as the heuristic function h (state) = number of squares from state to G assuming there were no barriers. For example, h (K) = 2 and h(S) = 4.
c. Hill-Climbing Search. Use the same heuristic function as in (b).
d. A* Search. Use the same heuristic function as in (b). Remove repeated states.
e. Is h an admissible heuristic? Prove your answer.
f. Is h2 (state) = min (2; h (state)) an admissible heuristic? Prove your answer.

4. [10 marks] You have a constraint satisfaction problem with the variables {A,B,C,D,E}, each having domain {1, 2, 3}. The problem has the following constraints.

a. $A \neq B$
b. $B \neq C$
c. $D \geq E$
d. $C \leq E$
e. $2 * A > E$
f. $B \leq E$

Apply Backtracking Search with Forward Checking and Arc Consistency to solve the Constraint Satisfaction Problem. Assume that Backtracking Search selects variables and values in increasing alphabetic/numerical order. Add into the following table how the domains of the variables change after each assignment. Circle

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values that are assigned by Backtracking Search, leave-out values that are removed by Forward Checking, and cross-out values that are removed by Arc Consistency. Stop when you found the first solution.

5. [4 marks]
   a. Which statement characterizes a rational agent best?
      - An agent that always selects the best action under the given circumstances.
      - An agent that performs reasoning steps to select the next action.
      - An agent that can explain its choice of an action.
      - An agent that is capable of predicting the actual outcome of an action.

   b. Which of the following factors is the most important one concerning space complexity for those search algorithms discussed in class that have exponential space complexity?
      - the number of nodes already visited
      - the number of nodes kept in the fringe
      - the average cost of moving from one node to the next
      - the memory footprint of a node (i.e. the memory required to store information about each node)

   c. What constitutes a full solution to a constraint satisfaction problem?
      - the set of all variables used in the specification of the problem
      - the set of all variables together with their respective potential values
      - an assignment of values to some (but not all) variables
      - an assignment of values to all variables

   d. In a knowledge base, a set of sentences which is called axiom is a
      - sentence that is taken as given without being derived from other sentences
      - sentence that is expressed in a language called a knowledge representation language
      - mechanisms that allow the system to learn for itself.



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Question 1: Assume that you are given the set of labeled training examples below, where each of three features has three possible values: a, b, or c. You choose to learn a decision tree from this data.

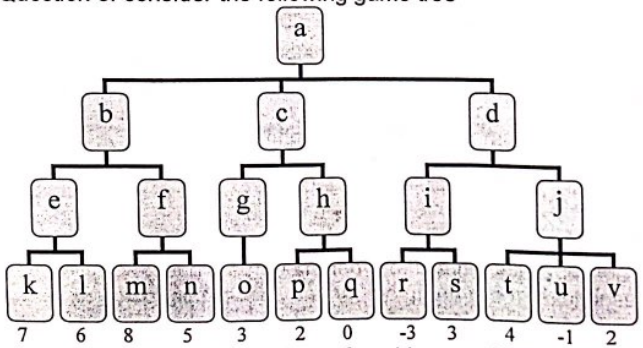
	F1	F2	F3	Output
ex1	c	b	b	+
ex2	a	a	c	+
ex3	b	c	c	+
ex4	b	c	a	-
ex5	a	b	c	-
ex6	c	a	b	-

a) [5 points] What score would the information gain calculation assign to feature F1?
b) [10 points] Which feature to be used as the root node for a decision tree being built?

Question 2:

a) [4 points] In best first search algorithm, what do a node's g-value and h-value represent?
b) [4 points] What does it mean to say that a heuristic is admissible, and what is the difference between admissibility condition and the optimality property?
c) [4 points] Prove the Breadth-first search, depth-first search are special cases of Best-first search.
d) [4 points] a best first search for f(n)=g(n) +h(n) where g(n) is the cost to reach the node n and h(n) is a heuristic function from the node n to the goal. Now, given that the graph is a tree, choose special functions g(n) and h(n), as general as possible, that will make best first search become
   a. Breadth-first search,
   b. Depth- first search

Question 3: consider the following game tree



[9 points] The First player to perform his move is the maximizing player and the scores are all from his point of view. Using alpha-beta search discuss what move should be chosen next?

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Question 4: [15 points] Consider the n-queens problem, which requires the placement of n queens on an n x n chess board such that no queen is on the same row, column, or diagonal as any other.

A. (9 pts) Formalize the 4-queens problem as a constraint satisfaction problem. (You only need to exhaustively list the constraints for one of the types of constraints that you include in the CSP. Any additional types of constraints can be explained briefly in words rather than equations.)
B. (4 pts) Show the constraint graph for your CSP formalization from part 1.
C. (2 pts) Name the uninformed search algorithm that is typically employed to solve CSPs?

Question 5: [5 points] Chose *the one* answer that best answers the question

1) In which of the following search algorithms is it possible to confuse a local maximum with a global maximum?
   a. depth first
   b. best first search
   c. hill climbing
   d. greedy search
2) An optimal search algorithm
   a. finds the solution that has the lowest path cost among all solutions
   b. finds all solutions
   c. finds a solution using the least amount of memory
   d. is guaranteed to find a solution when there is one
3) What is the main difference between the minimax and alpha-beta methods?
   a. only minimax is guaranteed to find the best move for Max
   b. alpha-beta pruning only works for randomly ordered successor nodes
   c. alpha-beta pruning typically examines fewer nodes than minimax
   d. minimax only works if the whole search tree can be computed before the respective best moves for Max and Min are determined
4) What is the key distinguishing characteristic of a learning agent?
   a. A learning agent can make a distinction between world states.
   b. A learning agent maps a real number (level-of happiness) to each state.
   c. A learning agent knows how to relate an action to a goal.
   d. A learning agent tries to improve its performance by suggesting improvements after observing its own actions and their outcomes.
5) What constitutes a full solution to a constraint satisfaction problem?
   a. the set of all variables used in the specification of the problem
   b. the set of all variables together with their respective potential values
   c. an assignment of values to some (but not all) variables
   d. an assignment of values to all variables

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Cairo University
Faculty of Science
Depart. Of Math.

Time allowed: 2 Hours
May, 2012
Instructor: Dr. Alaa Hosni

Final Examination of Artificial Intelligence-C408

Answer the following four questions:

QUESTION ONE [20 marks]

a) Consider the following 8-puzzle problem:

2	8	3
1	6	5
7		4

Initial state

1	2	3
8		4
7	6	5

Goal state

Consider the following heuristic function "the sum of all distances by which the tiles are out of place". Apply Best-first search algorithm.

b) Consider the following context-free grammar:

1. sentence ↔ noun_phrase verb_phrase
2. noun_phrase ↔ noun
3. noun_phrase ↔ article noun
4. noun_phrase ↔ article adjective noun
5. verb_phrase ↔ verb
6. verb_phrase ↔ auxiliary_verb verb
7. verb_phrase ↔ verb noun_phrase
8. article ↔ the
9. adjective ↔ brown
10. verb ↔ gone
11. noun ↔ dog
12. auxiliary-verb ↔ has

Define the transition network for this grammar, and then use it to parse the statement " the brown dog has gone"

QUESTION TWO [10 marks]

Consider the following game: nine matches are placed on a table. Between 2 opponents; at each move, the player must divide a pile of matches into two nonempty piles with different numbers of matches in each pile. The first player who can no longer make a move loses the game. Apply the minimax procedure illustrating the possible wins for Max. Suppose that MIN starts the first move. Do you think that MAX can force the game to win.

QUESTION THREE

a) Write the predicate `decimal` that converts a binary number into its decimal equivalent. Assume that the binary number is represented as a list of bits. [5 marks]

b) Write the Prolog predicat `objectiveFunction` that calculates the objective value for the 8-puzzle game, where the objective function is defined as the number of misplaced tiles. Your predicate should have the following form: [5 marks]

objectiveFunction (<currentState>, <goalState>, <functionValue>).

c) Consider the following program: [10 marks]

```
int(1).          int(2).          int(3).          int(4).
max(G,Gout):-int(N), N > G, max(N,Gout).
max(G,G).
```

Write all possible answers for the following goal:

```
goMax:-max(1,M),write(M),fail.
```

QUESTION FOUR [10 marks]

Trace the process of unifying the following two expressions:

parent(Y, father(Y), mother(john)) and parent(john, father(john), mother(Z))

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Department of Mathematics
Comp 408 / September 2013
Two pages

1. Answer with a conversed reasons
   For each of the following assets, assign a low, moderate, or high impact level for the loss of confidentiality, availability, and integrity, respectively. Justify your answers.
   a. [2 marks] An organization managing public information on its Web server.
   b. [2 marks] A law-enforcement organization managing extremely sensitive investigative information.
   c. [2 marks] A financial organization managing routine administrative information (not privacy-related information).
   d. [2 marks] An information system used for large acquisitions in a contracting organization that contains both sensitive, pre-solicitation phase contract information and routine administrative information. Assess the impact for the two data sets separately and the information system as a whole.
   e. [4 marks] A power plant contains a SCADA (supervisory control and data acquisition) system controlling the distribution of electric power for a large military installation. The SCADA system contains both real-time sensor data and routine administrative information. Assess the impact for the two data sets separately and the information system as a whole.

2. [14 marks] give a full proof of your answer
   Alice and Bob agree to communicate privately via e-mail using a scheme based on RC4, but they want to avoid using a new secret key for each transmission. Alice and Bob privately agree on a 128-bit key k. To encrypt a message m consisting of a string of bits, the following procedure is used.
   1. Choose a random 80-bit value v
   2. Generate the ciphertext $c = \text{RC4}(v \| k) \oplus m$
   3. Send the bit string $(v \| c)$
   a. Suppose Alice uses this procedure to send a message m to Bob. Describe how Bob can recover the message m from $(v \| c)$ using k.
   b. If an adversary observes several values $(v_1 \| c_1), (v_2 \| c_2), \ldots$ transmitted between Alice and Bob, how can he/she determine when the same key stream has been used to encrypt two messages?

3. [12 marks]
   Consider a very simple symmetric block encryption algorithm in which 32-bits blocks of plaintext are encrypted using a 64-bit key. Encryption is defined as

$$C = (P \oplus K_0) \boxplus K_1$$

   where C = ciphertext, K = secret key, K_0 = leftmost 64 bits of K, K_1 = rightmost 64 bits of K, $\oplus$ = bitwise exclusive OR, and $\boxplus$ is addition mod 2^{64}.
   a. Show the decryption equation. That is, show the equation for P as a function of C, K_0, and K_1.
   b. Suppose and adversary has access to two sets of plaintexts and their corresponding ciphertexts and wishes to determine K. We have the two equations:

$$C = (P \oplus K_0) \boxplus K_1; C' = (P' \oplus K_0) \boxplus K_1$$

   First, derive an equation in one unknown (e.g., K_0). Is it possible to proceed further to solve for K_0?

4. [12 marks] give a full steps proof of your answer
   Use an encryption algorithm to construct a one way hash function. Consider using

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RSA with a known key. Then process a message consisting of a sequence of blocks as follows: Encrypt the first block, XOR the result with the second block and encrypt again, and so on. Show that this scheme is not secure by solving the following problem. Given a two-block message B1, B2, and its hash, we have

$$\text{RSAH}(B1, B2) = \text{RSA}(\text{RSA}(B1) \oplus B2)$$

Given an arbitrary block C1, choose C2 so that RSAH(C1, C2) = RSAH(B1, B2). Thus, the hash function does not satisfy weak collision resistance.

5. [10 marks] give a detailed answer of the following:
In a public-key system using RSA, you intercept the ciphertext $C = 10$ sent to a user whose public key is $e = 5, n = 35$. What is the plaintext M?

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Cairo University
Faculty of Science
Depart. Of Math.

Time allowed: 2 Hours
September, 2013
Instructor: Dr. Alaa Hosni

Final Examination of Artificial Intelligence-C408

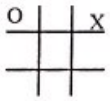
Answer the following four questions:

1.I) [5 marks] Algorithm verify_sentence that takes as argument a candidate expression and returns success if the expression is a sentence.

II) [5 marks] Convert the following wff to clause form:
$(\forall X)\ \{ N(b, X) \Rightarrow [\ M(b, X) \vee (\exists Y)\ (M(b,Y) \wedge N(Y, X)\ )\ ]\ \}$

III) [5 marks] Consider the following two clauses:
$\sim AT(john, Y) \vee AT(tom, Y)$, AT(john, home)
Show by resolution refutation that $(\exists X)$ AT(tom, X)

2.a) [10 marks] Consider the tic-tac-toe game. Define a suitable heuristic function, and then apply the minimax procedure to the following state (Suppose that MAX, represented by X has the next move)



b) [5 marks] write algorithm best-first search .

3) [marks] Consider the following context-free gramer:

1. sentence ↔ noun_phrase verb_phrase
2. noun_phrase ↔ noun
3. noun_phrase ↔ article noun
4. noun_phrase ↔ article adjective noun
5. verb_phrase ↔ verb
6. verb_phrase ↔ auxiliary_verb verb
7. verb_phrase ↔ verb noun_phrase
8. article ↔ the
9. adjective ↔ brown
10. verb ↔ gone
11. noun ↔ dog
12. auxiliary-verb ↔ has

a) [7 marks] Show that the following sentence can be derived using the given grammar: The brown dog has gone (write the rule number for each step).

b) [8 marks] Draw the transition network for the given grammar .

4. I) [5 marks] The following predicate sorts a list of numbers (for example the goal sort ([2,4,6,1,3,5], [], L), returns the answer L=[1,2,3,4,5,6])

```
sort(L, Li, Lo):- L=[H|T],!,findMax( T, H, Max), delete(L,Max, Loo), sort(Loo, [Max|Li], Lo).
sort([], L, L).
```

Write the predicates findMax that returns the maximum element in a list.

II) [10 marks] Consider the following prolog program.

```
in(p0, kitchen).    in(c1, diningroom ).   in(c2, kitchen).   in(c3, kitchen).   in(c4, kitchen).
plate(p0).
cup(c1).            cup(c2).               cup(c3).           cup(c4).
clean(p0).          clean(c2).             clean(c4).
dirty(c1).          dirty(c3).
```

What is the output of each of the following queries?

a. toWash:-in(X, kitchen), cup(X),dirty(X),write(X),fail.
b. toWash1:-in(X, kitchen), cup(X), !,dirty(X),write(X), fail.
c. toWashAgain:-in(X, kitchen), cup(X),clean(X),write(X),fail.

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Cairo University
Faculty of Science
Depart. Of Math.

Time allowed: 2 Hours
May, 2014
Instructor: Dr. Alaa Hosni

Final Examination of Artificial Intelligence-C408

- Check that you have 8 pages.

Answer the following four questions:

QUESTION ONE

[15 marks] Read the following programs and provide the answer to the query (only the first answer).

a.
```
p_1(X,[X, _ ]).
p_1(X,[ _ | L]):-p_1(X,L).

Query ?- p_1(X,[a,b,c,d]).
```

b.
```
p_2(L,R) :- acc(L, 0, R).
acc([_],A,A).
acc([H|T],A,R):- A1 is A+H, acc(T,A1,R).

Query ?- p_2([1,2,3,4,5],R).
```

c.
```
p_3([X|_], 1, X).
p_3([_|Xs], K, Y):- K > 1, K1 is K-1,   p_3(Xs,K1,Y).

?- p_3([a,b,c,d],3,R).
```

d.
```
p_5(nil, []).
p_5(t(X,nil,nil), [X]):- !.
p_5(t(X,L,R), Y):- p_5(L,NewL), p_5(R,NewR), append([X|NewL],NewR,Y).

Query ?- p_5( t(1, t(2, t(4, nil, nil),t(5, nil, nil)),t(3, nil, nil)), Z).
```

e.
```
p1(1).
p2(X) :- p1(X),!,fail.
p2(_).
p3(X) :- p1(X),fail,!.
p3(_).

Query ?- p2(1),p3(2).
```

a. .....

.....

b. .....

.....

c. .....

.....

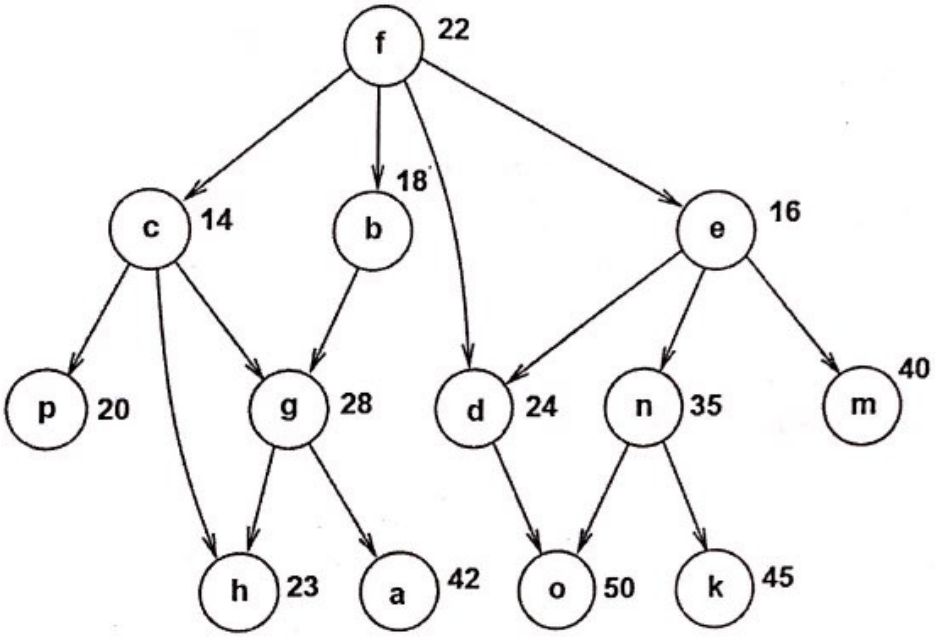
d. .....

.....

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QUESTION TWO

[15 marks] Consider the following state space, where each node represents a state, and each directed link represents a successor. Suppose the state space is to be searched, starting at node "f". List the nodes of the state space in the order that they are expanded for each of the following search methods. Assume that the search considers child nodes from left to right. You may assume that h(n) = the number to the right of node n, and that h(goal state)=100.



Breadth First Search:

.....

.....

Depth First Search:

.....

.....

gradient search

.....

.....

Best-first search

.....

.....

.....

.....

.....

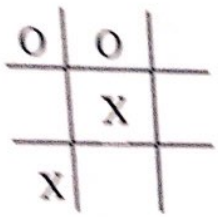
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QUESTION THREE

a. [8 marks] Apply algorithm unify to the following expressions. Either show their unifer or explain why they will not unify: family(X, mother(X), father(X)) and family (Y, mother(amr), father(father(adel))) .

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7 marks] Consider the following state.



Define a suitable heuristic function, and then apply the minimax procedure to the given state. Suppose that MAX(represented by X) has the next move.

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QUESTION FOUR

a. [5 marks]Find the composition $S_3\ S_2\ S_1$ where $S_1 = \{X/Y, W/Z\}$, $S_2 = \{V/X\}$, $S_3 = \{g(a)/V, b/W\}$

b. [5 marks] Consider the following piece of knowledge:

Tony and Mike belong to the Alpine club.
Every member of the Alpine club who is not a skier is a mountain climber.
Anyone who does not like snow is not a skier.

Represent this knowledge as a set of wffs.

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Cairo University
Faculty of science
Mathematics Department
120 minutes.

Final Exam (Spring 2015)
Artificial Intelligence
Comp 408

Answer the following questions. The total credit is 60.

Question (1) (16 points)

A. **(2 points)** How can you define Artificial Intelligence as "acting humanly"?

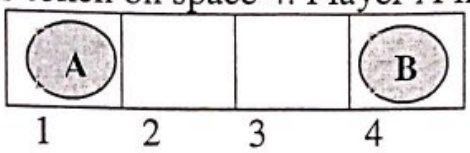
B. **(2 points)** What is a rational agent? And what is bounded rational agent?

C. **(4 points)** Define the PEAS (Performance Measure, Environment, Actuators, Sensors) of an internet shopping agent.

P.T.O.

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D. **(8 points)** Consider a two-player game; a board with four locations, from 1 to 4, in a line. Each player has a single token. Player A starts with his token on space 1, and player B starts with his token on space 4. Player A moves first.



initial state: State (1,4)

The two players take turns moving. Each player can move his token to an open adjacent space in either direction, left or right; or jump over the opponent to the next space if any. (For example, if A is on 3 and B is on 2, then A may move back to 1.) The game ends when one player reaches the opposite end of the board. If player A reaches space 4 first, then the value of the game is +1; if player B reaches space 1 first, then the value of the game is -1.

a. Draw the complete game tree, using the following conversions:
- Write each state as (S_A, S_B) where S_A and S_B denote the token locations.
- Put the terminal states in square boxes, and write its game value.
- Put loop states (states that already appear on the path to the root) in circles. Since it is not clear how to assign values to loop states, give them a question mark "?"

b. Evaluate the value of each node propagating upward from the leaves to the root. Explain in wards how you can handle the "?" value, and why.

c. Explain why the standard minimax algorithm would fail on this game tree and explain how you might fix it.

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Question (2) (16 points)

A. **(3 points)** What is the need for Skolem function in First Order Predicate Calculus?

B. **(7 points)** Explain how does the Genetic Algorithm work? Why is it considered as Local Search Algorithm?

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C. **(6 points)** Consider the following 8-puzzel states:

1	2	3
8		4
7	6	5

1	2	3
8	6	
7	5	4

Given a puzzle state like the one on the right, where the numbers are in the wrong places, we want to search for a series of moves which ends in the solution on the left. A path cost could be calculated as the number of moves required.

a. Write down a heuristic function, h, for this game (remember that this estimates the path cost from a particular state to the solution).

b. What is the value of h for the above board state?

c. In a greedy search, what move would be chosen next? (i.e., move the 3, the 4 or the 6 into the gap?) Is this heuristic admissible?

d. Can you suggest a better heuristic measure?

P.T.O.

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Question (3) (16 points)

A. **(8 points)** Suppose there are two friends living in different cities on a map. On every turn, we can move each friend simultaneously to a neighboring city on the map. The amount of time needed to move from city i to neighbor j is equal to the road distance d(i; j) between the cities, but on each turn the friend that arrives first must wait until the other one arrives (and calls the first on his phone) before the next turn can begin. We want the two friends to meet as quickly as possible.

a. What is the state space? Write a formal description of the state to use it in the rest of the question?

b. What is the successor function?

c. What is the goal state?

d. What is the cost function?

e. Let SLD(i; j) be the straight-line distance between any two cities i and j. Which, if any, of the following heuristic functions are admissible? Explain the reason of your choice?

(i) SLD(i; j) (ii) 2*SLD(i; j) (iii) SLD(i; j)/2 (iv) None

P.T.O.

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B. (8 points) Just write True/False:

i. (……..) The omniscience of an intelligent agent requires the agent to have infinite knowledge.

ii. (……..) Depth-first search always expands at least as many nodes as A* search with an admissible heuristic.

iii. (……..) True $\models$ False.

iv. (……..) Percept based intelligent agent collects information from its Actuators.

v. (……..) $(A \leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable

vi. (……..) Forward Chaining with the Generalized Modus Ponens is complete in First order predicate Calculus.

vii. (……..) A Unifier is a substitution for variables in atomic sentences that makes them identical

viii. (……..) An existentially quantified sentence with one variable can be instantiated several times as long as the Skolem constants used are all distinct.

Question (4) (16 points)

A. **(4 points)** Which of the following pairs can be unified and which can not. For those which can be unified, find a suitable substitution; For those which can not , explain why they cannot be unified.

(First pair)

likes(homer, dinner(x)) $\wedge$ (today(a) $\rightarrow$ dinner(x)).

likes(homer, dinner(cooked_by(y))) $\wedge$ (today(z) $\rightarrow$ dinner(cooked_by(Sally))).

(Second pair)

$p(x) \vee (q(f(x)) \wedge r(y))$.

$p(x) \vee (q(x) \wedge r(M))$.

B. **(6 points)** Translate the following sentences into first-order logic, using a function to represent mother:

All dogs are mammals

Fido is a dog

Fido's mother is a mammal

All mammals have a mother who is a mammal

P.T.O.

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C. **(6 point)** Consider the following Knowledge Base
Jack owns a dog. Every dog owner is an animal lover. No animal lover kills an animal. Either Jack or Curiosity killed the cat, whose name is Tuna.
Did Curiosity kill the cat?

a. Write this KB and your goal in First Order predicate calculus sentences

b. Put them CNF Form

c. Use Resolution-Refutation to answer the question

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Cairo University
Faculty of science
Mathematics Department
120 minutes.

Final Exam (Spring 2016)
Artificial Intelligence
Comp 408

Answer the following questions. The total credit is 60. **The Exam is in seven Pages**

Question (1) (17 points)

A. **(3 points)** Define an agent. Is any agent rational? What does it mean to be autonomous agent?

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B. **(4 points)** The Towers of Hanoi is the following problem:
There are three sticks A,B, and C. And there three disks on stick A. The top has diameter of 1, the middle disk has a diameter of 2, and the bottom 3. No disks on sticks B and C. You should move them all in the same order to C. You must move one disk at time and you cannot place a larger disk on a smaller one.

i. Find a representation of the problem that can be easily programmed.

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ii. Use your representation in (i) to describe the initial state , the goal state, and the successor function

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C. **(10 points)** Consider a state space where the **initial** state is number 1 and the successor function for state n returns two states, numbers 2n and 2n + 1. (Ex. The successors of state 1 are states 2 and 3)

i. Draw the portion of the state space for states 1 to 15.

ii. Suppose the goal state is 11. List the order in which nodes will be visited for breadth-first search, depth-limited search with limit 3, and iterative deepening search.

iii. Would bidirectional search be appropriate for this problem? If so, describe in detail how it would work.

iv. What is the branching factor in each direction of the bidirectional search?

v. Does the answer to (iii) suggest a reformulation of the problem that would allow you to solve the problem of getting from state 1 to a given goal state with almost no search?

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Question (2) (15 points)

A. **(2 points)** Show that the following sentence in propositional logic is valid.
$(p \rightarrow Q) \leftrightarrow (\sim Q \rightarrow \sim P)$

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B. **(8 points)** Circle T or F for True/False: **(two circles is a wrong answer)**

i. (T / F) Iterative Deeping Search Algorithm is optimal.
ii. (T / F) An agent is usually operating in a partially observable environment.
iii. (T / F) A* Search cannot be used in robotics because percepts, states, and actions are continuous.
iv. (T / F) Depth-first Iterative deepening always returns the same solution as breadth-first search if b is finite.
v. (T / F) The clause $(\sim A \vee \sim B \vee C)$ entails the clause $(\sim A \vee \sim B)$
vi. (T / F) A Sound inference rule can derive all sound sentences from a knowledge base
vii. (T / F) Context-free grammars may fail to capture English grammar because the meaning of an English sentence may depend on the context of the utterance.
viii. (T / F) Ambiguity in natural language can always be resolved by checking dictionary definitions of words

C. **(3 points)** What affects the speed of the genetic algorithm?

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D. **(2 points)** Is Forward chaining complete in first-order logic? Explain why?

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Question (3) (15 points)

i. **(7 points)** Represent the following knowledge base in first-order logic. Use the predicates: attend(x), fair(t), pass(x,t), prepared(x), smart(x), study(x), cu_std(x).

a. Everyone who is smart, studies, and attends class will be prepared.
b. Everyone who is prepared will pass a test if the test is fair.
c. Every Cairo University student i.e. cu_std is smart.
d. If a test is not fair, no one will pass the test
e. Hany is a cu_student
f. At least one student passed the AI test
g. Hany attends class.

ii. **(4 points)** Convert this knowledge base into conjunction normal form, CNF.

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iii. **(4 points)** Use resolution to prove that:

Study(Hany) → pass (Hany, AI)

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Question (4) (13 points)

A. **(7 points)** Consider the game of 2×2 tic tac toe (X-O) where each player has the additional option of passing (i.e. making no action). Assume O (The player Min) goes first.

i. Draw the full game tree down to depth 2. (first Min's Turn, then Max's Turn)
ii. Suppose the evaluation function is the number of Xs minus the number of Os. Mark the values of all leaves Then propagate upward using Minimax fashion.
iii. Circle any node that would not be evaluated by alpha-beta pruning.
iv. Suppose we want to solve the game to find the optimal move. Explain why apha-beta would be better than minimax.

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B. **(6 points)** Consider the sentence "the man walked slowly to the supermarket" and consider the following lexicon and grammars:

Noun -> man | supermarket
V -> walked
Adv -> slowly
Prep -> to
Det -> the

A

S	->	NP VP
NP	->	Noun
NP	->	Article NP
VP	->	Verb Vmod
Vmod	->	Adv Vmod
Vmod	->	Adv
Adv	->	PP
PP	->	Prep NP

B

S	->	NP VP
NP	->	Noun
NP	->	Article NP
VP	->	Verb Adv
Adv	->	Adv Adv
Adv	->	PP
PP	->	Prep NP

C

S	->	NP VP
NP	->	Noun
NP	->	Article Noun
VP	->	VP PP
VP	->	VP Adv Adv
VP	->	Verb
PP	->	Prep NP

i. Which of the three grammars, combined with the lexicon, generate the given sentence.
ii. Show the corresponding parse tree.
iii. Write down one sentence of <u>non-English</u> generated by one of the above grammars. (Explain how it is generated by the grammar)

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Cairo University
Faculty of science
Mathematics Department
120 minutes.

Final Exam (Spring 2017)
Artificial Intelligence
Comp 408

Answer the following questions. The total credit is 60. **The Exam is in seven Pages**

Question (1) (15 points)

A. **(5 points)** When would you describe an agent to be intelligent? and when would it be rational? Give an example of an agent that is neither intelligent nor rational. Then explain how you would change it to be rational.

B. **(3 points)** Describe the PEAS of the MARS robot, and Characterize its operating environment (observable or not, deterministic or not,...etc)

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C. **(7 points)** Explain why the depth first algorithm is not complete while the breadth first is. How would you improve the depth to be complete.

Question (2) (15 points)

A. **(1 point)** What is the definition of an admissible heuristic?

B. **(4 points) Multiple Choice:**

1. Which of the following algorithms is optimal
   a. Bidirectional
   b. Iterative deepening Algorithm
   c. Hill Climbing
   d. Both (a) and (b)
   e. None of the above
2. $\exists x(S(x) \wedge C(x))$ is equivalent to which of the following?
   a. $\exists x(C(x) \wedge S(x))$
   b. $\exists x S(x) \wedge \exists x C(x)$
   c. $\sim\forall x(S(x) \rightarrow \sim C(x))$
   d. Both (a) and (c)
   e. All the above

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3. How could you translate "There are at most two apples" into FOL
   a. $\forall x \forall y \forall z$ ((Apple(x) $\wedge$ Apple(y) $\wedge$ Apple(z)) $\rightarrow$ (x=y $\vee$ y=z $\vee$ x=z))
   b. $\forall x \forall y$ (Apple(x) $\wedge$ Apple(y)) $\rightarrow$ (x=y $\vee$ y=x)
   c. $\exists x \exists y$ (Apple(x) $\wedge$ Apple(y) $\wedge \sim(x = y) \wedge \forall z$ (Apple(z) $\rightarrow (x = z \vee y = z)$)
   d. Both (a) and (c)
   e. None of the above

4. How could you translate "There are exactly two apples" into FOL
   a. $\exists x \exists y$ ((Apple(x) $\wedge$ Apple(y) $\wedge \forall z$ Apple(z)) $\rightarrow$ (x=z $\vee$ y=z))
   b. $\exists x \exists y$ (Apple(x) $\wedge$ Apple(y) $\wedge \sim(x = y) \wedge \forall z$(Apple(z) $\rightarrow$ (x=z$\vee$y=z)))
   c. $\exists x \exists y$ ($\sim(x = y) \wedge \forall z$(Apple(z) $\leftrightarrow$ (x=z $\vee$ y=z)))
   d. $\exists x \exists y$ (Apple(x) $\wedge$ Apple(y)$\wedge \sim(x = y)) \wedge \forall x \forall y \forall z$((Apple(x) $\wedge$ Apple(y) $\wedge$ Apple(z)) $\rightarrow$ (x=y $\vee$ y=z $\vee$ x=z))
   e. Both (b) and (c) and (d)
   f. All the above

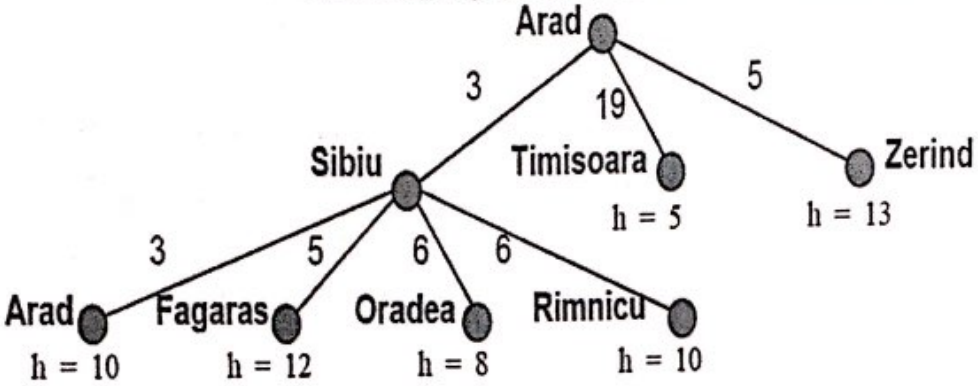
5. **(8 points)** Just T for True or F for False:
   i. (T / F) No Intelligent agent can behave in a partial observable environment.
   ii. (T / F) The depth limited Search Algorithm is optimal.
   iii. (T / F) Alpha-beta pruning with a good heuristic evaluation function yields an optimal playing strategy
   iv. (T / F) The simulated annealing can fall into local maxima
   v. (T / F) "p and q are not both true" is equivalent to " p and q are both not true"
   vi. (T / F) Any valid statement in first-order logic can be proved via propositionalization.
   vii. (T / F) The statement " Most birds can fly " can be expressed in FOL.
   viii. (T / F) Context-free grammars may fail to capture English grammar, because the meaning of an English sentence may depend on the context of the utterance.

6. **(2 points)** In the following, Skolemize to eliminate the existential quantifier
   a. $\exists x\ course(x) \wedge$ take (Ahmed, x)
   b. $\forall x[(\exists y\ Animal\ (y) \wedge \sim Loves(x, y)) \vee (\exists z\ Loves(z, x))]$

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Question (3) (15 points)

A. (7 points) The following diagram shows a partially expanded search tree. Each arc is labeled with the corresponding step cost, and the leaves are labeled with the h values.



a. Which leaf will be expanded next by an A* search?

b. Which leaf will be expanded next by a greedy search?

c. If the goal can be reached only from Fagaras at the fourth level. Which blind algorithm BFS or DFS will reach the goal first. Explain why?

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B. **(8 points)** Suppose a genetic algorithm uses chromosomes of the form x = abcdefgh with a fixed length of eight digit genes. Each gene can be any digit between 0 and 9. Let the fitness function of individual x be calculated as:

$$f(x) = (a + b) - (c + d) + (e + f) - (g + h)$$

and let the initial population consists of four individuals with the following chromosomes

x1 = 65413532 x2 = 87126601
x3 = 23921285 x4 = 41852094

a. Evaluate the fitness of each individual, and arrange them in order with the fittest first.

big first

b. Perform a crossover on the fittest two individuals using one point crossover at the middle point

c. By looking at the fitness function, find the chromosome representing the optimal solution (i.e. with maximum fitness value). Find the value of the maximum fitness.

d. By looking at the initial population of the Algorithm, can you say whether it will be able to reach the optimal solution without applying the mutation operation? Why ?

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Question (4) (15 points)

A. (9 points) Resolution:

"Tony and Sally belong to ABC club. Every member of the ABC club is either a skier (متزحلق) or mountain climber(متسلق جبال) or both. No mountain climber likes rain and all skiers like snow(جليد) . Sally does not like whatever Tony likes and likes whatever Tony does not like. Tony likes rain and snow"

Prove using resolution and unification that "Sally is a mountain climber but not a skier"

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B. (6 points) NLP: Consider the following review

"I bought my wireless keyboard/mouse set several months ago, and like a lot of products, it has some unpleasant issues. On the good side is the styling. The design is fresh, clean, and interesting. The keyboard can tilt at different angles, which was important because I had some difficulty typing with it flat. The mouth and the keyboard proved durable and reliable despite a number of mishaps. "

Each inference below is based on a possible understanding of the review. For each inference, briefly explain whether that inference was primarily based on syntactic, semantic, pragmatic, discourse, or world knowledge

a. The reviewer owns the keyboard.
b. The word "like" in the first line means "similar to" (i.e. does not mean love)
c. The charger is part of the keyboard.
d. The reviewer had difficulty typing with the keyboard.
e. The reviewer likes the keyboard.
f. The reviewer paid money for that keyboard set

Best Wishes
Areeg Abdalla

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